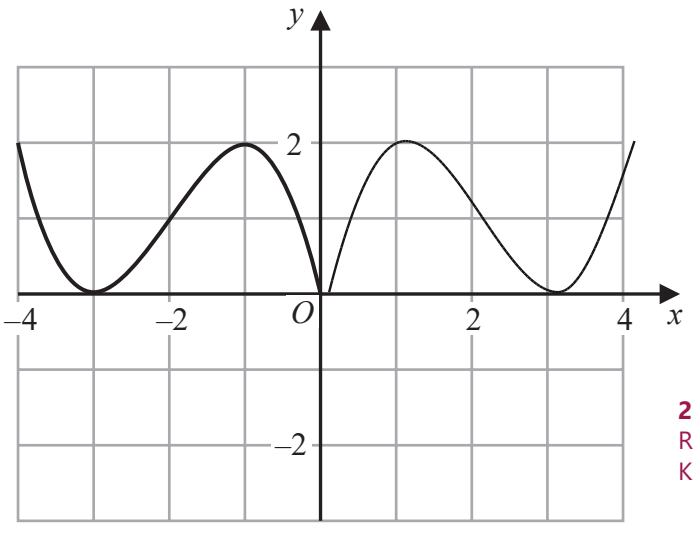
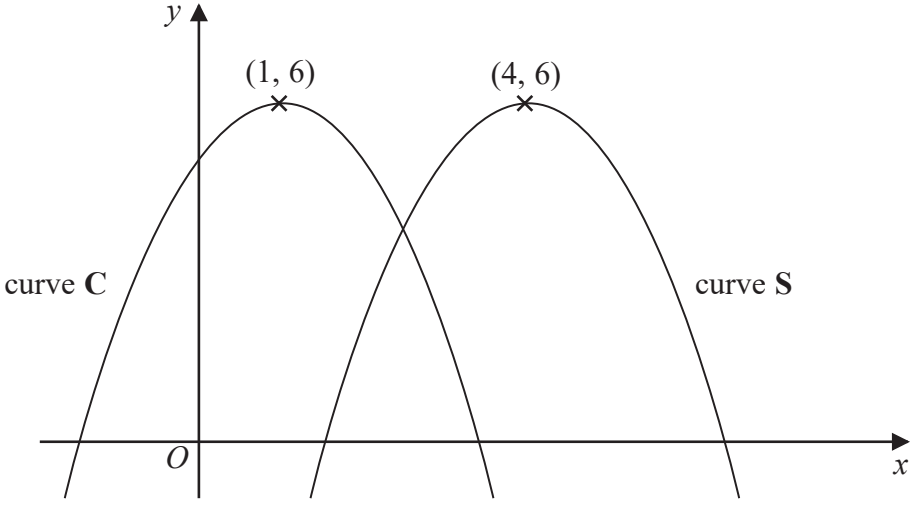
21 The graph of the curve with equation $y = f(x)$ is shown on the grid below.



2 marks total
Reflect original in y-axis
Key reflected points

(a) On the grid above, sketch the graph of the curve with equation $y = f(-x)$

(2)



The curve **C** with equation $y = 5 + 2x - x^2$ is transformed by a translation to give the curve **S** such that the point (1, 6) on **C** is mapped to the point (4, 6) on **S**.

(b) Find an equation for **S**.

Translation is 3 right: replace x by (x – 3).

S: $y = 5 + 2(x - 3) - (x - 3)^2$.

2 marks total
3 right: f(x–3)
Correct equation for S

(2)

(Total for Question 21 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA